

Supp. Fig. 3c — MHC class II extension

Reviewer round 2, minor point 1 — “repeat this analysis using a larger benchmark set”

For verification before the response goes out. Generated by `make_review_pdf.py`; every number read from the tracked CSVs.

What changed. The published panel was 16 class I post-training-cutoff triads. It now carries 10 class II as well. The class I points are untouched — same CSV, same transform, same colour, same statistics, enforced by a regression check (`plot_supp3c.py --check`) that fails rather than publish a moved result.

Five of the ten needed no new inference. They were already folded in the published `pdb.nf` run for Figure 1; Supp. Fig. 3c had simply been drawing from a validation table that is class I only. Their CDR RMSD was already computed (`cdr_rmsd_af3_0`, seed 0, exactly as the class I points were), so only PTI-PAE needed extracting — which reads existing predictions and needs no GPU. That extraction was verified by running `8enh` and `8es9` through it alongside: both reproduce their published values to the last digit.

Two were excluded on chemistry. `8trl` and `8trr` are citrullinated epitopes (α -enolase and vimentin on HLA-DR4). Citrulline has no standard one-letter code, so RCSB canonicalises it to arginine and AF3 folds a different chemistry from the deposit — and for these epitopes the modification is the point, since DR4’s shared-epitope pocket repels arginine and it is citrullination that creates the neo-epitope. Both were post-cutoff, already folded, and predicted well (3.34 and 1.22 Å), so they are omitted on chemistry rather than on quality, consistent with the rule applied elsewhere in the project. Removing them strengthened every statistic: class II Spearman went from -0.61 to -0.73 and the combined Spearman from $p = 0.008$ to $p = 0.001$.

Five were newly folded through the pipeline’s own `COMPOSE_INFERENCE_JSON` and `INFERENCE` (MSAs on, 5 seeds, 1 diffusion sample), then measured with the same four processes `pdb.nf` uses for its own RMSDs.

PDB	released	CDR RMSD (Å)	PTI-PAE	peptide	MHC	on TSV	prediction
8pjj	2024-06-19	1.54	3.54	PKYVKQNTLKLAR	DRA*01:01	yes	published run
8vcy	2024-08-07	1.94	11.90	GQVELGGSSPETCI	DQA1*03:01	yes	published run
9yag	2026-08-19	2.01	4.51	QFPFPQEPFPC	DQA1*05:01	yes	newly folded
9aud	2025-02-05	2.04	5.05	QVYSLIRPNENPA	H2-Aa	yes	newly folded
9yaf	2026-08-19	2.09	5.79	AQFPFPQELPYPG	DQA1*05:01	yes	newly folded
8vcx	2024-08-07	2.17	8.74	GQVELGGGPGAESCQ	DQA1*03:01	yes	published run
8vq8	2025-01-29	2.44	6.54	QVYSLIRPNENPA	H2-Aa	yes	newly folded
8vd0	2024-08-07	3.02	10.40	GQVELGGGNAVEVCKG	DQA1*03:01	yes	published run
8vd2	2024-08-07	3.44	12.20	GQVELGGGTPIESCQ	DQA1*03:01	yes	published run
27eb	2026-08-26	22.17	17.52	APQPEQPFPWQC	DQA1*05:01	NO	newly folded

All 10 are post-training-cutoff. AF3’s cutoff is 2023-01-12, taken from `clean_pdb.py`, which cites the AF3 paper’s data-availability statement. The earliest release here is `8pjj` at 2024-06-19 — 524 days past it. This is distinct from `max_template_date`, which defaults to 2021-09-30 and caps what templates AF3 may retrieve, so no structure here could have been handed its own answer.

	n	Pearson r	p	Spearman ρ	p
MHC class I	16	-0.53	0.036	-0.60	0.014
MHC class II	10	-0.75	0.012	-0.73	0.016
combined	26	-0.50	0.010	-0.59	0.001

Class II is significant on its own and stronger than class I, on an independent set.

Three things to check

1. `27eb` is not on the TSV — we derived it from RCSB rather than taking it from a published listing. See below.
2. `27eb` is the outlier, and the most informative point. It is the least confident (PTI-PAE 17.5) and least accurate (22.2 Å) triad in the class II set — AF3 docks it badly and PTI-PAE flagged it in advance. It sits with `8gom/8gon`, the class I reverse-dock failures, which is the reviewer’s question answered in a single point. It was flagged as suspicious on confidence before its RMSD was computed. All five of its chains matched deposited entities exactly and its crystal copy was chosen on measured contact geometry, so the input is sound — but it is the one triad without a published-listing provenance, and so the point a reviewer is most likely to question.
3. Two copies per asymmetric unit in `9yaf`, `9yag` and `27eb`. Chains were not taken by index order — the published

table has **8tr1** pairing MHC **A/B** with TCR **I/J**, so index order is not reliably copy order. Each candidate was checked against deposited coordinates, and every selected chain sits 4–7 Å from the MHC α chain.

Supp. Fig. 3c, extended

CDR RMSD vs AF3 PTI-PAE on post-training-cutoff cognate triads. Blue is the published 16 class I points, unchanged; red is the 12 class II points added here. Dashed lines are per-class fits — pooling them would assert a relationship the combined statistics support only weakly, since the two classes sit at different absolute PTI-PAE. Regenerated by `plot_supp3c.py --check`.

class I: n=16	$r=-0.53$ ($p=0.036$)	$\rho=-0.60$ ($p=0.014$)
class II: n=10	$r=-0.75$ ($p=0.012$)	$\rho=-0.73$ ($p=0.016$)
all: n=26	$r=-0.50$ ($p=0.010$)	$\rho=-0.59$ ($p=0.001$)

